

Since r is True and v is False, by defn. of bi-conditional, $r \leftrightarrow v$ is False. This is a contradiction to ③.

$\therefore q$ is False. From ②, and defn. of bi-conditional, $\neg u$ is False.

By defn. of negation $\neg(\neg u)$ is True. From the bi-conditional $\neg(\neg u) \leftrightarrow u$ being True, u is True. $\therefore$ John did the access $\rightarrow$ Sure

Let $r, \neg s$ be True. By defn. of negation, s is False. From ③, and defn. of bi-conditional, v is True. By defn. of negation, $\neg v$ is False. By defn. of AND, $(\neg r \wedge \neg v)$ is False. By defn. of bi-conditional, $s \leftrightarrow (\neg r \wedge \neg v)$ is True, which matches ④.

If r is True, i.e. Carlos tells the truth, then Diana did the access

But this is a contradiction to the fact that exactly one person must access.

$\therefore r$ is False. From ③, and defn. of bi-conditional, v is False.

$\therefore s$ is True (From given constraints).

By defn. of negation, $\neg r$ and $\neg v$ are each ^{True} ~~False~~. By defn. of AND, $s \leftrightarrow (\neg r \wedge \neg v)$ is True. which is consistent with ④.

$\therefore$ Ans: John did the access

b) Constraint: Exactly one is lying.

Suppose, p is True. From ⑤ and defn. of biconditional, $\neg s$ is True. By defn. of conditional, From ③, v is True. $\therefore$ Diana did the access

By defn. of negation, s is False. By defn. of ~~negation~~ negation, $\neg v$ is False. By defn. of AND, $(\neg r \wedge \neg v)$ is False. $\therefore$ By defn. of bi-conditional, $s \leftrightarrow (\neg r \wedge \neg v)$ is True (matches with ④)

s is False $\rightarrow$ Diana lies.

$\therefore p$ is True $\rightarrow$ Alice tells the truth. By defn. of biconditional and ①, t is True. $\therefore$ Carlos did the access. From constraint, only one person accesses. This is a contradiction.